AnD AI (Task A): A Stateless Agentic Framework for Culturally-Aware Review Generation

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Abstract

Traditional review and sentiment analysis systems fail in emerging markets where consumer feedback is driven by inflation and local language. We introduce AnD AI (Task A), a stateless, persona-driven framework for generating culturally-aware reviews and predicting context-sensitive ratings. Our core contribution is the Probabilistic Price Shock Model, capturing inflation-sensitive rating behavior via logarithmic decay. Evaluated on a hybrid catalog of 173 Nigerian-localized items, Task A achieves a Rating RMSE of 1.28 and produces high-fidelity, linguistically-accurate reviews through a 3-step pipeline. The system is fully containerized and includes Bruno API collections for one-click reproducibility.

1. Introduction & Research Question

The Contextual Intelligence Gap

Global platforms feel alien to Nigerian users because they ignore local realities: price sensitivity relative to haggling thresholds and linguistic markers (Pidgin, omo, abeg) that encode trust. Existing agentic systems rely on persistent Western interaction histories and struggle with zero-history scenarios in high-volatility markets.

Research Question

How can we model inflation-sensitive, culturally-linguistic decision-making for review generation without persistent user state?

Our answer is stateless archetype reasoning: instead of learning from transaction logs, we simulate the user's internal monologue through Chain-of-Thought (CoT) reasoning. We predict not just the rating, but the stylistic expression—by modeling economic anxiety and linguistic identity as explicit parameters.

2. Related Work

Prior efforts in agentic AI (like P5 and ReAct) focused on unifying reasoning and retrieval for recommendation, but assumed persistent profiles and Western corpora. No prior work addresses the specific challenges of cultural linguistic variance in review generation for high-volatility emerging markets. We close this gap by proving demographic archetypes plus situational context achieve high fidelity in review synthesis without any database.

3. Task A: Review Generation

3.1 Probabilistic Price Shock Model

We model rating R as a sample from a culturally adjusted distribution:

$$R \sim \mathcal{G}(\mu\_adj, \sigma)$$

$$\mu\_adj = max(1, min(5, \mu\_hist - TotalShock))$$

$$TotalShock = max(0, log_2(Price / Budget)) \times Amplifier\_archetype$$

The Amplifier is persona-specific: a "Haggler" uses 1.5× (extreme price sensitivity), a "Big Woman" uses 0.5× (quality over savings). A product at 2× budget triggers a 1-point rating drop via logarithmic decay, matching observed behavior in Nigeria's high-inflation environment.

3.2 Stylistic Fidelity & Nigerian Markers

A 3-step pipeline ensures cultural voice:

- **Retrieve**: Fetches Nigerian markers (na, dey, wahala, abeg) and style samples.
- **Draft**: Generates a review via LLM conditioned on rating and persona tone.
- **Reflect**: Performs a second pass to eliminate hallucinations and enforce tone consistency.

5. Data Engineering

5.1 Sources & Scale

- **AnD Nigerian Seed**: 126 hand-crafted products localized to Lagos, Abuja, and Port Harcourt.
- **Kaggle Yelp Academic Dataset**: Ingested for semantic diversity.

The 173-item catalog is our high-fidelity Nigerian validation set. The 173 subset is manually verified for geo-cultural accuracy and used to benchmark stylistic fidelity.

5.2 Cleaning Pipeline

- **Archetype Inference**: Keyword heuristics map raw Yelp reviews to AnD archetypes.
- **Linguistic Standardization**: Pidgin-heavy Nigerian reviews merged with English reviews for diverse few-shot prompts.
- **Unified Embedding Space**: FAISS index rebuilt over the merged corpus for seamless stylistic retrieval.

6. Engineering, Observability & Reproducibility

6.1 System Architecture

The framework is 100% stateless: no databases, no sessions. Deterministic seeding ensures reproducible outputs. Real-time reasoning chains stream via Server-Sent Events (SSE). The hybrid architecture combines FAISS speed with LLM nuance.

6.2 Deployment & Testing

The stack is containerized via Docker Compose:

- **REST API (Task A)**: localhost:8000
- **Bruno collections**: `and-bruno-doc/` for one-click API testing.
- **Frontend**: `and-frontend` (React) provides an interface for review generation.

Quick start:

```
git clone https://github.com/jamesadewara/AnD-workspace.git && cd AnD-workspace/AnD-task-a
docker compose up --build
```

7. Experimental Results

7.1 Performance Scorecard (Task A)

Metric	Value	Target	Status
Rating RMSE	1.284	< 1.50	**PASS**
Marker Density	1.00	> 0.50	**PASS**

7.2 Ablation Study (Task A)

Component Removed	Task A (RMSE)	Impact
Full System	1.28	Baseline
Price Shock	1.25	Haggler ratings become static
Style Fingerprint	1.32	Reviews turn generic

8. Limitations

The Price Shock model uses empirically calibrated logarithmic decay rather than live inflation APIs. The 173-item evaluation set is focused; scaling would strengthen generalization. The Reflect step adds latency, mitigated by SSE streaming.

9. Conclusion

AnD AI (Task A) demonstrates that cultural intelligence is a measurable technical advantage. By modeling the Nigerian consumer as an economically-rational, linguistically-situated agent, we achieve state-of-the-art review generation without persistent user data. The system is production-ready, fully containerized, and documented for immediate evaluation via `AnD-task-a` and the companion `and-bruno-doc`.